

Topic: Logic

Part 1

Logical Expressions

- Section 1 Introduction
- Section 2 Statements
- Section 3 Logical Operators
- Section 4 Compound Statements

Part 2

Truth Tables and Logical Equivalence

- Section 1 Introduction
- Section 2 Order of Operations
- Section 3 Truth Tables
- Section 4 Tautology and Contradiction
- Section 5 Logical Equivalence

Part 3

Conditional Statement and Its Variations

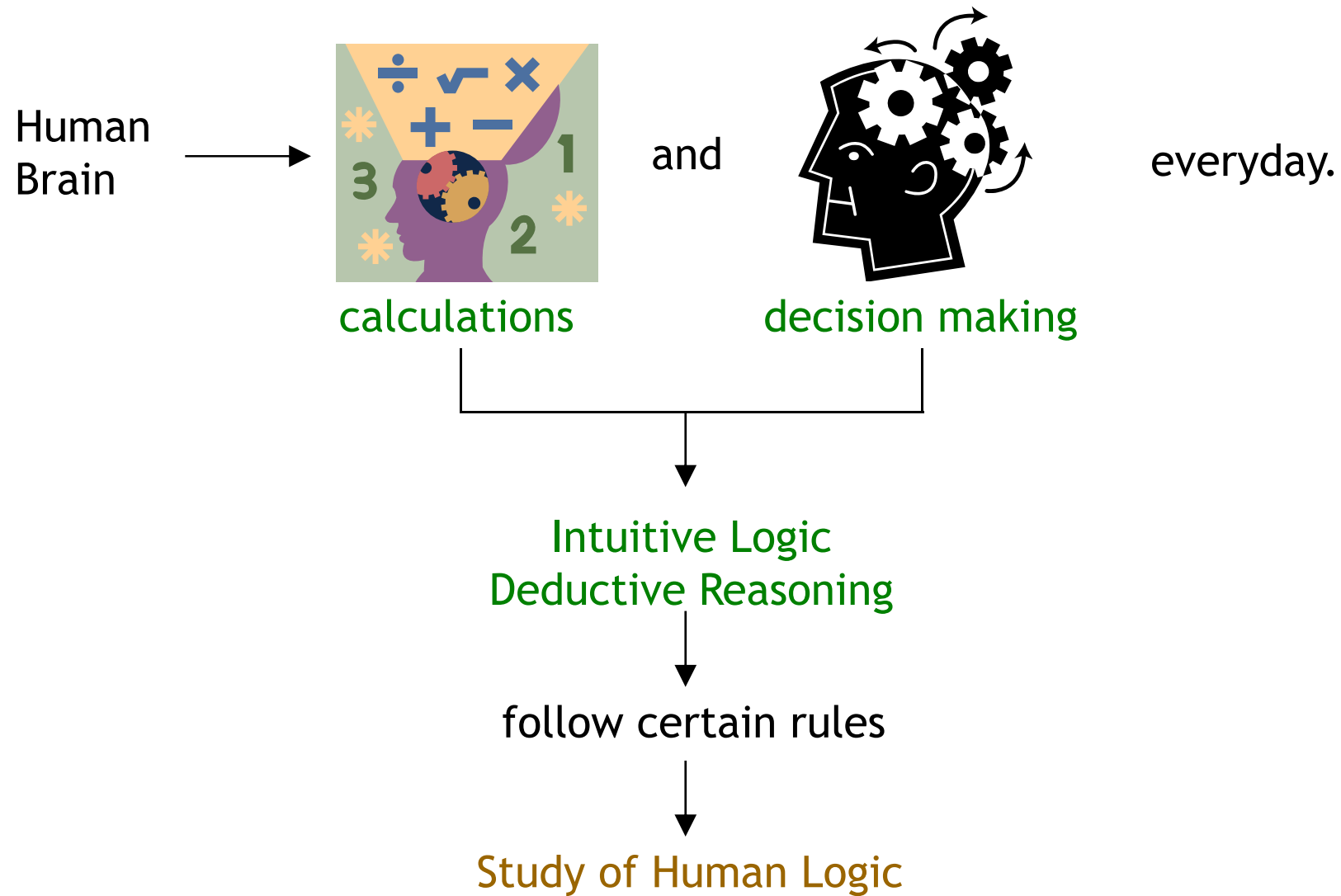
- Section 1 Introduction
- Section 2 Conditional Statement
- Section 3 Contraposition, Converse, Inverse Statements

Part 1 Logical Expressions

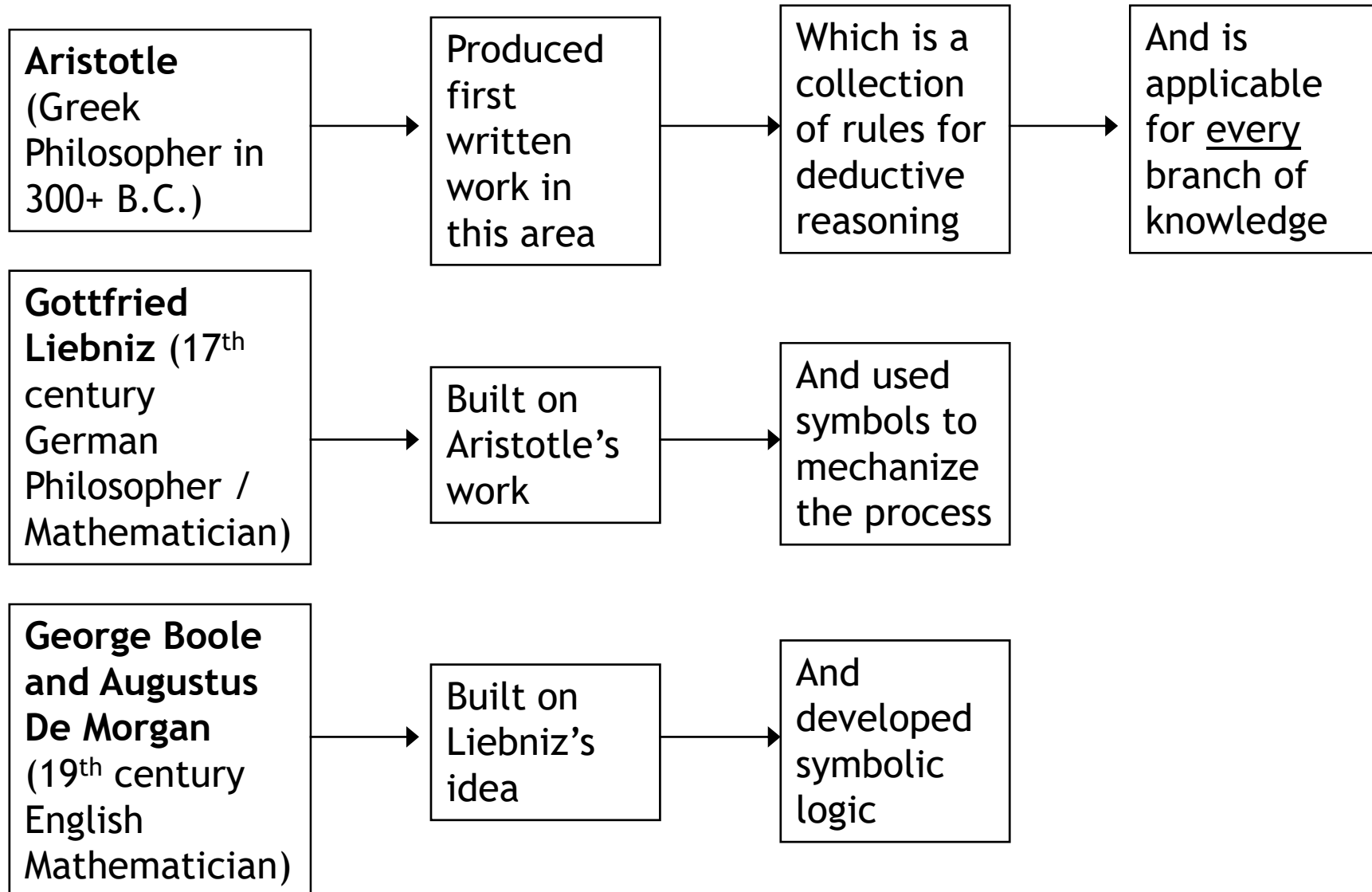
Section 1 Introduction

- Introduction
- Objective
- The Different Parts of a Logical Expression

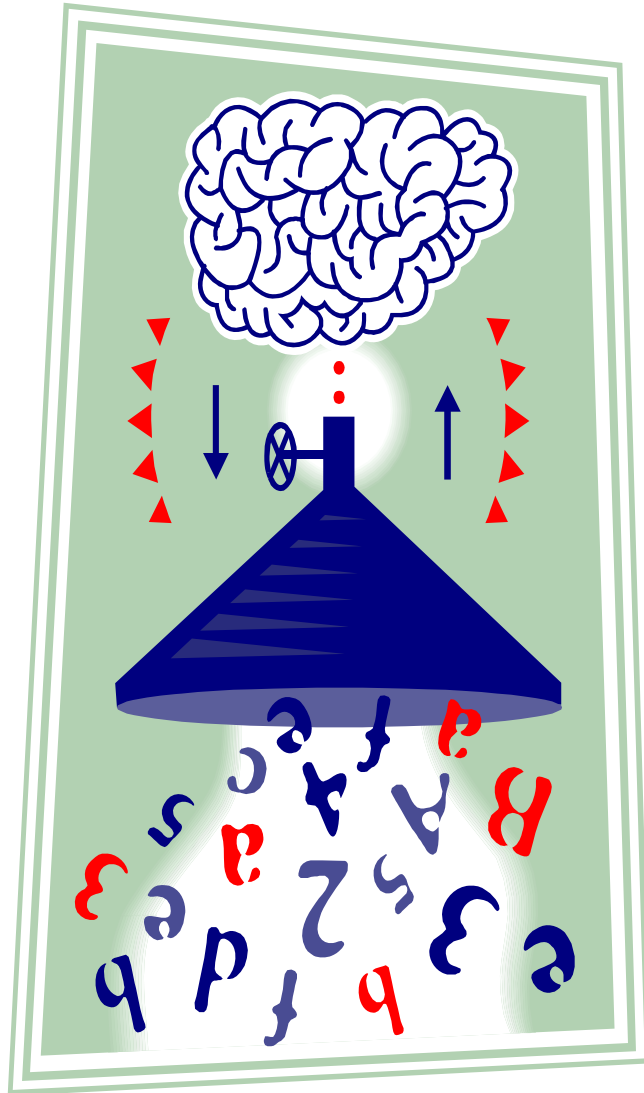
How Logic Came About



History Xpress



What is symbolic Logic?



Intuitive Logic
of Human Brain

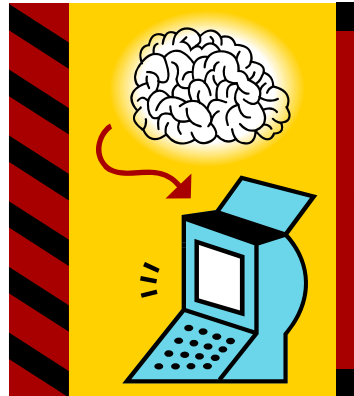
Replicate the series of
decision making we make

A systematic method for
describing and simplifying
the logic processes.

Logic Processes

Symbolic Logic

Symbolic Logic Today



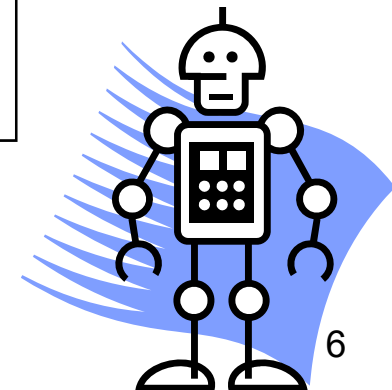
**Symbolic
Logic**

It is the
theoretical
basis used
in
**computer
science**

And is used areas like

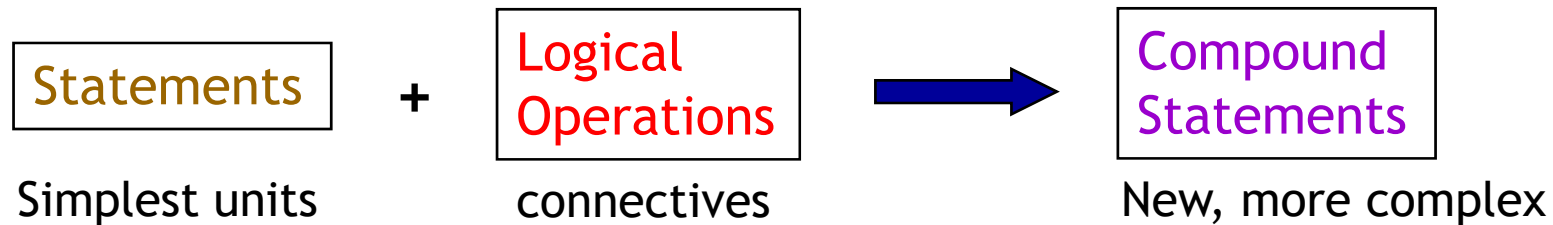
1. Digital Logic Circuit Design
2. Relational Database Theory
3. Automata Theory
4. Artificial Intelligence

Research
is still on-
going



Different Parts of the Logical Expressions

- To construct **logical expressions** using **symbols**.



Part 1 Logical Expressions

Section 2 Statements

- Introduction
- Statements
- Statement Variables
- Truth Values

Statements

Definition

A statement is a sentence that is known to be either true or false, but not both.

Statement variables, usually denoted by small letters e.g. p or q, are used in the place of a statement for shorthand purposes.

This state of being true (**T**) or false (**F**) is called the truth value of the statement.

Example

Consider the following :

1. “ $1 + 1 = 2$ ”  → true statement

2. “ $2 + 2 = 5$ ”  → false statement

3. “ $x + y = 3$ ”
→ sometimes true
→ sometimes false → truth value ??? → not a statement

Part 1 Logical Expressions

Section 3 Logical Operators

- Introduction
- Logical Operator - Negation
- Logical Operator - Conjunction
- Logical Operator - Disjunction

Logical Operations

In symbolic logic, there are 3 logical operations :

$\sim$ means “not”

$\vee$ means “or”

$\wedge$ means “and”

These logical operations are used to connect statements together.

So given statement variables p and q,

$\sim p$, called **negation of p**, means **not p**.

$p \wedge q$, called **conjunction of p**, means **p and q**.

$p \vee q$, called **disjunction of p**, means **p or q**.

Logical Operations - Negation

Definition

The negation of a statement variable p , written as $\sim p$, means “not p ”, or the **opposite** of p .

It follows the rule that : If p is T, then $\sim p$ is F. If p is F, then $\sim p$ is T.

- The table for all possible truth values of p is represented below :

p	$\sim p$
T	F
F	T

- Hence, p and $\sim p$ have opposite truth values.
- The negation of a statement can be taken repeatedly.
- Thus $\sim\sim p = p$

That's like turning a switch from on to off (or vice versa).



Example (Logical Operations - Negation)

Consider the truth values of the following statements :

1. The sun sets in the west.

T

2. The sun does not set in the west.

F

3. $1+1 \neq 0$

T

4. $1+1 = 0$

F

Note

Statement 2 is the negation of statement 1. Hence, they have opposite truth values.

Statement 4 is the negation of statement 3. Hence, they have opposite truth values.

Logical Operations - Conjunction

Definition

The conjunction of 2 statement variables p and q , written as $p \wedge q$, means “ p and q ”.

- The table for all possible combinations of p and q is represented below :

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F



p

q



That's like having
a series circuit!

- $p \wedge q$ is true only if both p and q are true.
- Otherwise, it is false.

Example (Logical Operations - Conjunction)

Consider the truth values of the following statements :

1. The sun sets in the west AND $1+1 = 2$.

T

2. The sun sets in the west AND $1+1 = 0$.

F

3. The sun sets in the east AND $1+1 = 2$.

F

4. The sun sets in the east AND $1+1 = 0$.

F

Only this is true
since both
statement
variables are
true.

These are false
since one of the
two statement
variables is false.

Logical Operations - Disjunction

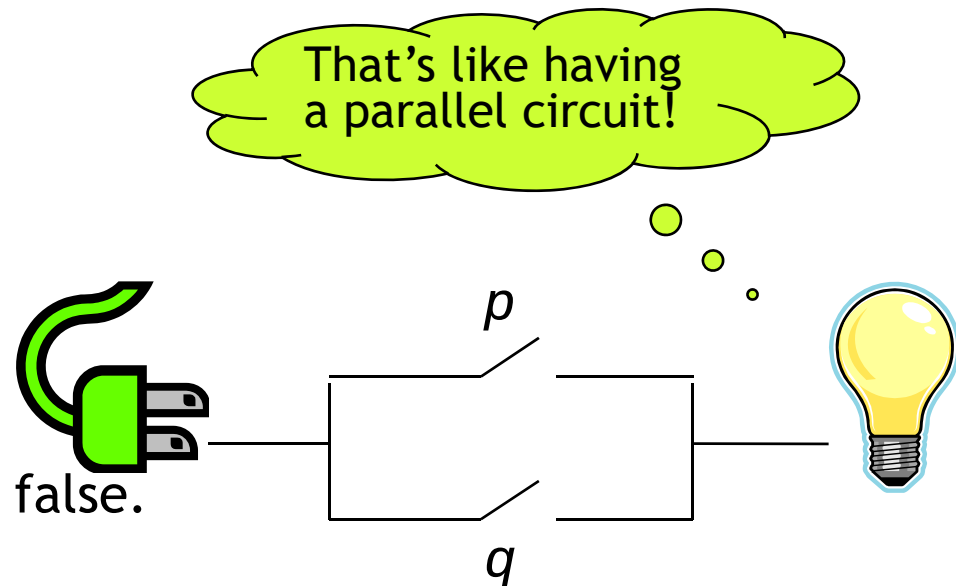
Definition

The disjunction of 2 statement variables p and q , written as $p \vee q$, means “ p or q ”.

- The table for all possible combinations of p and q is represented below :

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

- $p \vee q$ is false if both p and q are false.
- Otherwise, it is true.



Example (Logical Operations - Disjunction)

Consider the truth values of the following statements :

1. The sun sets in the west OR $1+1 = 2$.
2. The sun sets in the west OR $1+1 = 0$.
3. The sun sets in the east OR $1+1 = 2$.
4. The sun sets in the east OR $1+1 = 0$.

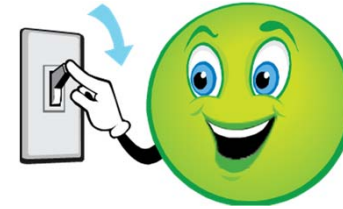
T
T
T
F

All these are true
since one of the
statement
variables is true.

Only this is false
since both the
statement variables
are false.

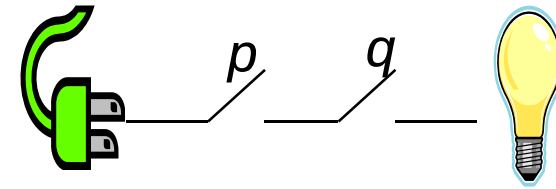
Summary of Logical Operators

p	$\sim p$
T	F
F	T



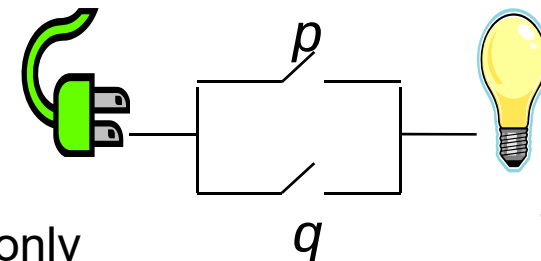
p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Will only light up here



p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Will NOT light up here only



Part 1 Logical Expressions

Section 4 Compound Statements

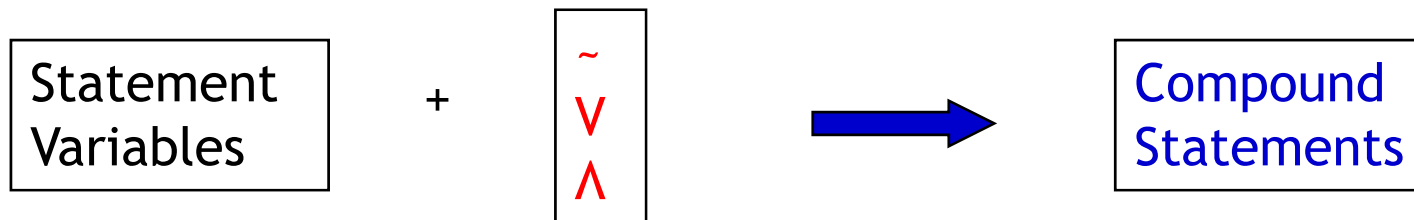
- Introduction
- Translating English To Symbols
- Translating Mathematics To Symbols

Compound Statements

Definition

Compound Statements are more complex statements built by incorporating statement variables (e.g. p , q) with logical operations ($\sim$, $\wedge$, $\vee$).

We denote compound statements using capital letters e.g. X and Y .



Example $X = p \vee (q \wedge r)$ or $Y = \sim (q \vee r) \wedge \sim p$

Translating English to Symbols

Special Cases To Take Note Of :

p but q means p and q
neither p nor q means $\text{not } p$ and $\text{not } q$

Example (Translating from English to Symbols)

Let h = “Ewen is hungry” and s = “Ewen is sleepy”.

Translate the following sentences to symbols :

a. Ewen is not hungry but he is sleepy.

$\equiv$ Ewen is not hungry and Ewen is sleepy.

$\equiv \quad \sim h \wedge s$

b. Ewen is neither hungry nor sleepy.

$\equiv$ Ewen is not hungry and Ewen is not sleepy.

$\equiv \quad \sim h \wedge \sim s$

Exercise (Translating from English to Symbols)

Let a = “Jon likes apples”, b = “Jon likes oranges”, c = “Jon likes grapes”.

Translate the following sentences to symbols :

a. Jon likes apples and oranges.

≡ Jon likes apples and Jon likes oranges.

≡ $a \wedge b$

b. Jon does not like grapes.

≡ It is not true that Jon like grapes.

≡ $\sim c$

c. Jon likes apples or oranges.

≡ Jon likes apples or Jon likes oranges.

≡ $a \vee b$

d. Jon likes apples but not grapes.

≡ Jon likes apples and Jon does not like grapes.

≡ $a \wedge \sim c$

Exercise (Translating from English to Symbols)

Let a = “Jon likes apples”, b = “Jon likes oranges”, c = “Jon likes grapes”.
Translate the following sentences to symbols :

e. Jon likes neither apples, nor oranges, nor grapes.

$\equiv$ Jon doesn't like apples and Jon doesn't like oranges and Jon doesn't like grapes.

$\equiv \quad \sim a \quad \wedge \quad \sim b \quad \wedge \quad \sim c$

f. Jon likes neither apples nor grapes, but he likes oranges.

$\equiv$ Jon doesn't like apples and Jon doesn't like grapes and Jon likes oranges.

$\equiv \quad \sim a \quad \wedge \quad \sim c \quad \wedge \quad b$

g. Jon likes grapes, and either apples or oranges.

$\equiv$ Jon likes grapes and (Jon likes apples or Jon likes oranges).

$\equiv \quad c \quad \wedge \quad (\quad a \quad \vee \quad b \quad)$

Translating Mathematics to Symbols

Special Cases To Take Note Of :

$x \leq a$	means	$x < a$ <u>or</u> $x = a$
$a < x < b$	means	$x > a$ <u>and</u> $x < b$

Example (Translating from Mathematics to Symbols)

Let $p = "x > 2"$, $q = "x < 5"$ and $r = "x = 5"$.

Translate the following mathematical sentences to symbols :

a. $x \neq 5$

$\equiv$ x is not equal to 5

$\equiv \sim r$

c. $2 < x < 5$

$\equiv$ x is more than 2 but less than 5

$\equiv x > 2$ and $x < 5$

$\equiv p \wedge q$

b. $x \leq 5$

$\equiv$ x is less than or equal to 5

$\equiv x < 5$ or $x = 5$

$\equiv q \vee r$

d. $2 < x \leq 5$

$\equiv$ x is more than 2 but
less than or equal to 5

$\equiv x > 2$ and $(x < 5$ or $x = 5)$

$\equiv p \wedge (q \vee r)$

Exercise (Translating from Mathematics to Symbols)

Let e = “sum > 20”, f = “sum < 50”, g = “sum = 20” and h = “sum = 50”.

Translate the following mathematical sentences to symbols :

a. $\text{sum} \neq 20$ or $\text{sum} \neq 50$

$$\equiv \sim g \vee \sim h$$

b. $\text{sum} \geq 20$

$\equiv$ sum is more than or equal to 20

$\equiv$ $\text{sum} > 20$ or $\text{sum} = 20$

$$\equiv e \vee g$$

c. $20 < \text{sum} < 50$

$\equiv$ sum is more than 20 but less than 50

$\equiv$ $\text{sum} > 20$ and $\text{sum} < 50$

$$\equiv e \wedge f$$

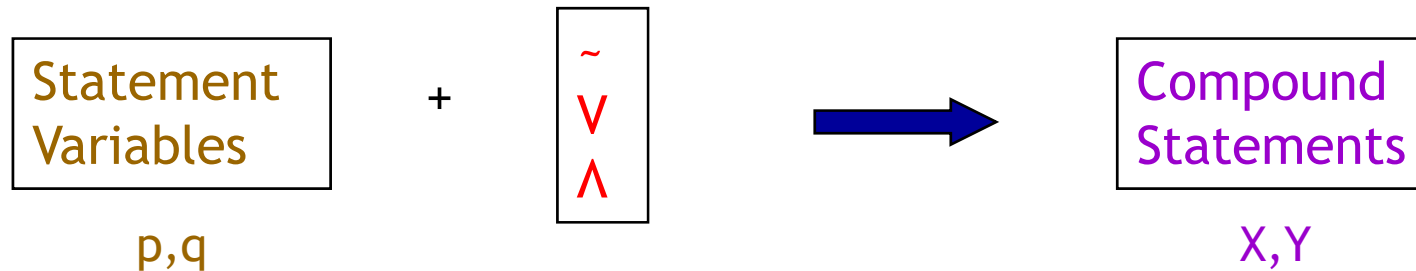
d. $20 \leq \text{sum} \leq 50$

$\equiv$ sum is more than or equal to 20 but less than or equal to 50

$$\equiv (\text{sum} > 20 \text{ or } \text{sum} = 20) \text{ and } (\text{sum} < 50 \text{ or } \text{sum} = 50)$$

$$\equiv (e \vee g) \wedge (f \vee h)$$

Summary



- Translate English $\rightarrow$ Symbols

p but q means p and q
neither p nor q means not p and not q

- Translate Mathematics $\rightarrow$ Symbols

$x \leq a$ means $x < a$ or $x = a$
 $a < x < b$ means $x > a$ and $x < b$

Part 2 Truth Tables and Logical Equivalence

Section 1 Introduction

- Introduction To Truth Tables
- Introduction To Logical Equivalence

Truth Tables and Logical Equivalence

- The truth value of a compound statement depends on the truth values of the statement variables.
- A simple and concise way to show the relationship between the statement variables and the compound statement involved is through the use of a [truth table](#).
- If two compound statements evaluate to the same truth values for all possible combinations of the statement variables, then they are [logically equivalent](#).

Part 2 Truth Tables and Logical Equivalence

Section 2 Order of Operations

- Introduction
- Example and Exercise

Order of Operations

Before going into the actual construction of truth table, it is important to know that in statements that involve $\sim$, $\wedge$ and $\vee$, the **order of operations** follow these rules :

1. Perform $\sim$ first

$p \wedge \sim q$ means $p \wedge (\sim q)$

2. $\wedge$ and $\vee$ are co-equal

So $p \vee q \wedge r$ would be ambiguous.
Therefore, always use ().
E.g $(p \vee q) \wedge r$ or $p \vee (q \wedge r)$

3. The use of brackets ()
will supersede all other
operations.

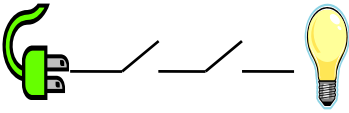
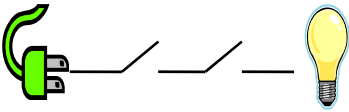
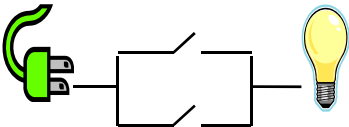
That's the same as
normal algebra!

Example (Order of Operations)

1. Perform $\sim$ first
2. $\wedge$ and $\vee$ are co-equal
3. $()$ supersedes all others

Given that $x = F$ and $y = T$, evaluate $(x \vee y) \wedge \sim(x \wedge y)$.

Solution :

$$\begin{aligned} & (x \vee y) \wedge \sim(x \wedge y) \\ = & \underbrace{(F \vee T)} \wedge \sim(\underbrace{F \wedge T}) \\ = & T \wedge \sim(\underbrace{F \wedge T}) \\ = & T \wedge \underbrace{\sim F} \\ = & \underbrace{T \wedge T} \\ = & T \end{aligned}$$


Exercise (Order of Operations)

Given that $x = T$, $y = T$ and $z = F$, evaluate $\sim(x \wedge y) \vee z$.

Solution :

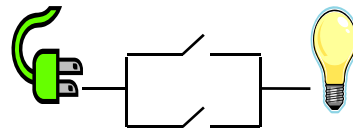
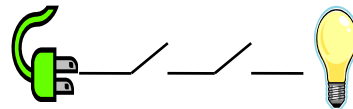
$$\sim(x \wedge y) \vee z$$

$$= \sim(\underbrace{T \wedge T}_{\text{true}}) \vee F$$

$$= \underbrace{\sim T}_{\text{false}} \vee F$$

$$= \underbrace{F \vee F}_{\text{false}}$$

$$= F$$



Part 2 Truth Tables and Logical Equivalence

Section 3 Truth Tables

- Definition and Objective
- How To Construct The Rows
- How To Build The Columns
- Example and Exercises

Definition of Truth Tables

A Truth Table is a listing of all possible arrangements of truth values of the statement variables (e.g. p and q) as well as the compound statements (e.g. $\sim p \vee \sim q$).

Objective

Show how *statement variables* (*inputs*) of a logic expression *relate to compound statement* (*output*).

Input				Output
p	q	r		$(p \vee q) \wedge r$
T	T	T	...	T
T	T	F	...	F
T	F	T	...	T
T	F	F	...	F
F	T	T	...	T
F	T	F	...	F
F	F	T	...	F
F	F	F	...	F

Left :
Inputs

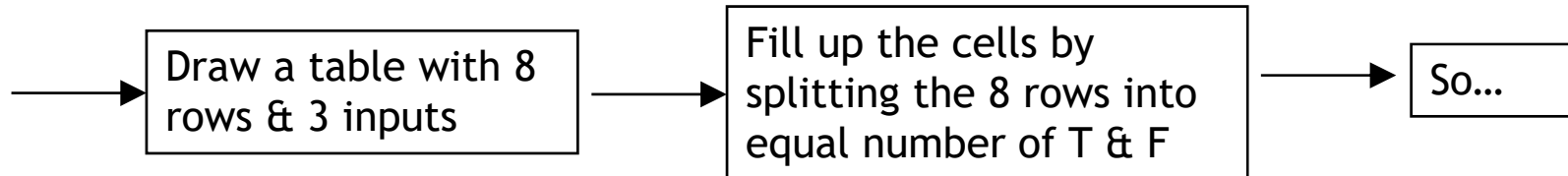
Right :
Output

How To Construct The Rows

- No. of rows = 2^N
where N is the no. of statement variables (inputs)
- 2^N rows will show all possible arrangements of the inputs.
- The next example provides a step by step guide which will guarantee
 - *all possible arrangements,*
 - *with no missing or duplicated rows.*

Example (How to Construct The Rows)

- Given : 3 inputs a, b, c
- No. of rows = $2^N = 2^3 = 8$



a	b	c
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

1st Input :
4T by 4F

2nd Input :
2T by 2F, then
repeat

3rd Input :
1T by 1F, then repeat

Exercise (How to Construct The Rows)

p	q	r	s
T	T	T	T
T	T	T	F
T	T	F	T
T	T	F	F
T	F	T	T
T	F	T	F
T	F	F	T
T	F	F	F
F	T	T	T
F	T	T	F
F	T	F	T
F	T	F	F
F	F	T	T
F	F	T	F
F	F	F	T
F	F	F	F

Given : 4 inputs p, q, r, s

No. of rows = 2^N

$$= 2^4 = 16$$

1st Input :
8 by 8

2nd Input :
4 by 4

3rd Input :
2 by 2

4th Input :
1 by 1

How To Build The Columns

- Build simpler components
- Then consolidate to get final output

Example (Truth Table)

Prepare the truth table for $(x \vee y) \wedge \sim(x \wedge y)$

No. of inputs = 2

No. of rows = $2^N = 2^2 = 4$

Step 1 : Set up Rows

1st input 2x2

2nd input 1x1

x	y				
(1)	(2)				
T	T				
T	F				
F	T				
F	F				

This row shows the
column number in ()

Example (Truth Table) (Cont'd)

Prepare the truth table for $(x \vee y) \wedge \sim(x \wedge y)$

=> Set up columns for the simpler components

x	y	$x \vee y$	$x \wedge y$	$\sim(x \wedge y)$	
(1)	(2)	(3)=(1) $\vee$ (2)	(4)=(1) $\wedge$ (2)	(5)= $\sim$ (4)	
T	T				
T	F				
F	T				
F	F				

Step 2 : Strategize

This row shows the column number in ()

Example (Truth Table) (Cont'd)

Prepare the truth table for $(x \vee y) \wedge \sim(x \wedge y)$

=> Set up columns for the simpler components

Step 2 : Strategize

Fill in each cell

x	y	$x \vee y$	$x \wedge y$	$\sim(x \wedge y)$	
(1)	(2)	(3)=(1) $\vee$ (2)	(4)=(1) $\wedge$ (2)	(5)= $\sim$ (4)	
T	T	T	T	F	
T	F	T	F	T	
F	T	T	F	T	
F	F	F	F	T	

This row shows the
column number in ()

Example (Truth Table) (Cont'd)

Prepare the truth table for $(x \vee y) \wedge \sim(x \wedge y)$

Consolidate $\Rightarrow$ Perform $\sim/\vee/\wedge$ to columns from Step 2

Step 3 : Consolidate

x	y	$x \vee y$	$x \wedge y$	$\sim(x \wedge y)$	Output
(1)	(2)	(3)=(1) $\vee$ (2)	(4)=(1) $\wedge$ (2)	(5)= $\sim$ (4)	(3) $\wedge$ (5)
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

This row shows the column number in ()

Exercise (Truth Table for 3 Inputs)

Prepare the truth table for $\sim(x \wedge y \wedge z) \vee ((x \vee z) \wedge (y \vee z))$

x	y	z	$x \wedge y \wedge z$	$\sim(x \wedge y \wedge z)$	$x \vee z$	$y \vee z$	$(x \vee z) \wedge (y \vee z)$	Output
(1)	(2)	(3)	(4)	(5)= $\sim$ (4)	(6)=(1) $\vee$ (3)	(7)=(2) $\vee$ (3)	(8)=(6) $\wedge$ (7)	(5) $\vee$ (8)
T	T	T	T	F	T	T	T	T
T	T	F	F	T	T	T	T	T
T	F	T	F	T	T	T	T	T
T	F	F	F	T	T	F	F	T
F	T	T	F	T	T	T	T	T
F	T	F	F	T	F	T	F	T
F	F	T	F	T	T	T	T	T
F	F	F	F	T	F	F	F	T

Part 2 Truth Tables and Logical Equivalence

Section 4 Tautology and Contradiction

- Tautology
- Contradiction
- Examples and Exercises

Tautology and Contradiction

Definition

A tautology t is a compound statement that is always true, regardless of the truth values of the statement variables.

A contradiction c is a compound statement that is always false, regardless of the truth values of the statement variables.

Example (Tautology)

Show that $p \vee \sim p$ is a tautology.

Step 1 : Set up Rows

Step 2 : Strategize

Solution

No. of inputs = 1 No. of rows = $2^1 = 2$

Input		Output
p	$\sim p$	$p \vee \sim p$
T	F	T
F	T	T



Since truth value = T for all rows, therefore output is a tautology.

Exercise (Contradiction)

Show that $p \wedge \sim p$ is a contradiction.

Step 1 : Set up Rows

Step 2 : Strategize

Solution

No. of inputs = 1 No. of rows = $2^1 = 2$

Input		Output
p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F



Since truth value = F for all rows, therefore output is a contradiction.

Part 2 Truth Tables and Logical Equivalence

Section 5 Logical Equivalence

- Definition
- Technique of Proof
- Proof of Logical Equivalence
- Proof of Logical Non-Equivalence

Definition

2 compound statements P and Q (outputs) are logically equivalent if and only if they evaluate to the same truth values for all possible combinations of the statement variables (inputs).

Logical equivalence is denoted by $P \equiv Q$.

Technique To Proof

Technique for Testing Whether 2 Compound Statements P and Q are Logically Equivalent

1. Construct a truth table, with one output column for P and one output column for Q.
2. For each combination of inputs, check the truth values of P and Q.
 - a. If the truth values of P and Q are the same for each and every row, then $P \equiv Q$.
 - b. If the truth values of P and Q are not the same for some rows, then P and Q are not logically equivalent.

Example (Proof of Logical Equivalence)

Prove that $\sim(\sim p) \equiv p$.

Solution

No. of inputs = 1 No. of rows = $2^1 = 2$

Input		Output (LHS)	Output (RHS)
p	$\sim p$	$\sim(\sim p)$	p
T	F	T	T
F	T	F	F

Since p and $\sim(\sim p)$ have same truth values for each and every row, they are logically equivalent.

Step 1 : Set up Rows

Step 2 : Strategize

Exercise (Proof of Logical Equivalence)

Prove or disprove $\underbrace{\sim(p \vee q)}_{\text{LHS}} \equiv \underbrace{\sim p \wedge \sim q}_{\text{RHS}}$.

Step 1 : Set up Rows

Step 2 : Strategize

Solution

No. of inputs = 2 No. of rows = $2^2 = 4$

Inputs			LHS			RHS
p	q	$p \vee q$	$\sim(p \vee q)$	$\sim p$	$\sim q$	$\sim p \wedge \sim q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Since every row has the same truth values, we have proved that $\sim(p \vee q) \equiv \sim p \wedge \sim q$.

Example (Logical Equivalence and Tautology)

If t is a tautology, show that $p \vee t \equiv t$.

Step 1 : Set up Rows

Step 2 : Strategize

Solution

No. of inputs = 1 (since 2nd input must = T) No. of rows = $2^1 = 2$

Inputs		Output
p	t	$p \vee t$
T	T	T
F	T	T



Since $p \vee t$ has same truth values as t for all rows, therefore $p \vee t \equiv t$.

Exercise (Logical Equivalence and Contradiction)

If c is a contradiction, show that $p \vee c \equiv p$.

Step 1 : Set up Rows

Step 2 : Strategize

Solution

No. of inputs = 1 (since 2nd input must = F) No. of rows = $2^1 = 2$

Inputs		Output
p	c	$p \vee c$
T	F	T
F	F	F



Since $p \vee c$ has same truth values as p for all rows, therefore $p \vee c \equiv p$.

Example (Proof of Logical Non-equivalence)

Prove or disprove $\underbrace{\sim(p \vee q)}_{\text{LHS}} \equiv \underbrace{\sim p \vee \sim q}_{\text{RHS}}$.

Step 1 : Set up Rows

Step 2 : Strategize

Solution :

No. of inputs = 2 No. of rows = $2^2 = 4$

Inputs			LHS			RHS
p	q	$p \vee q$	$\sim(p \vee q)$	$\sim p$	$\sim q$	$\sim p \vee \sim q$
T	T	T	F	F	F	F
T	F	T	F	F	T	T
F	T	T	F	T	F	T
F	F	F	T	T	T	T

Since rows 2 and 3 have different truth values,
 $\sim(p \vee q)$ and $\sim p \vee \sim q$ are not logically equivalent.

Exercise (Proof of Logical Non-equivalence)

Prove or disprove $\underbrace{\sim(p \wedge q)}_{\text{LHS}} \equiv \underbrace{\sim p \wedge \sim q}_{\text{RHS}}$.

Step 1 : Set up Rows

Step 2 : Strategize

Solution :

No. of inputs = 2 No. of rows = $2^2 = 4$

Inputs			LHS			RHS
p	q	$p \wedge q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$\sim p \wedge \sim q$
T	T	T	F	F	F	F
T	F	F	T	F	T	F
F	T	F	T	T	F	F
F	F	F	T	T	T	T

Since rows 2 and 3 have different truth values,
 $\sim(p \wedge q)$ and $\sim p \wedge \sim q$ are not logically equivalent.

So Far ...

- Truth Table - Need to know
 - How to construct rows → No. of rows = 2^N
 - How to construct columns

- Then Use Truth Table as a **tool** to prove
 - Tautology (**t**) & Contradiction (**c**)
 - Logical Equivalence and Non-equivalence

Part 3 Conditional Statement and Its Variations

Section 1 Introduction

- Conditional Statement
- Variations of Conditional Statement

Conditional Statement

- Conditional Statement is a statement with a hypothesis and a conclusion.
- Conditional Statement makes programming so much more powerful because it allows us
 - to *test* a variable against a value; and
 - to act in one way if the condition is met; or
 - to act in another way if the condition is not met.
- Hence, they are also known as the “if-then” statement.

Example of the Conditional Statement

Conditional Statement

If you work hard, then you will pass this course.

So, if you really meet the condition and work very hard (hypothesis is T) , then you will definitely pass this course (conclusion is T).

However, if you did not work hard (hypothesis is F), chances are, you will fail this course (conclusion is F).

Variations of the Conditional Statement

Many different statements can be made by :

- switching the hypothesis with the conclusion; and/or,
- using the negation of a statement instead of the original statement.

The three most common variations of conditional statement are :

1. Contrapositive Statement;
2. Converse Statement;
3. Inverse Statement.

As these statements can appear somewhat similar to the original conditional statement, a good understanding of these variations will reduce a lot of confusion in your programming work.

Part 3 Conditional Statement and Its Variations

Section 2 Conditional Statement

- Definition & Truth Table
- Rationale Behind the Truth Table
- Order of Operations
- Example - Truth Table Involving Conditionals
- Exercise - Logical Equivalence Involving Conditionals

Definition & Truth Table

Definition

Let p and q be statement variables.

A conditional $p \rightarrow q$, read as “ q by p ” means “if p then q ” or “ p implies q ”.

The truth table is as follows :

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

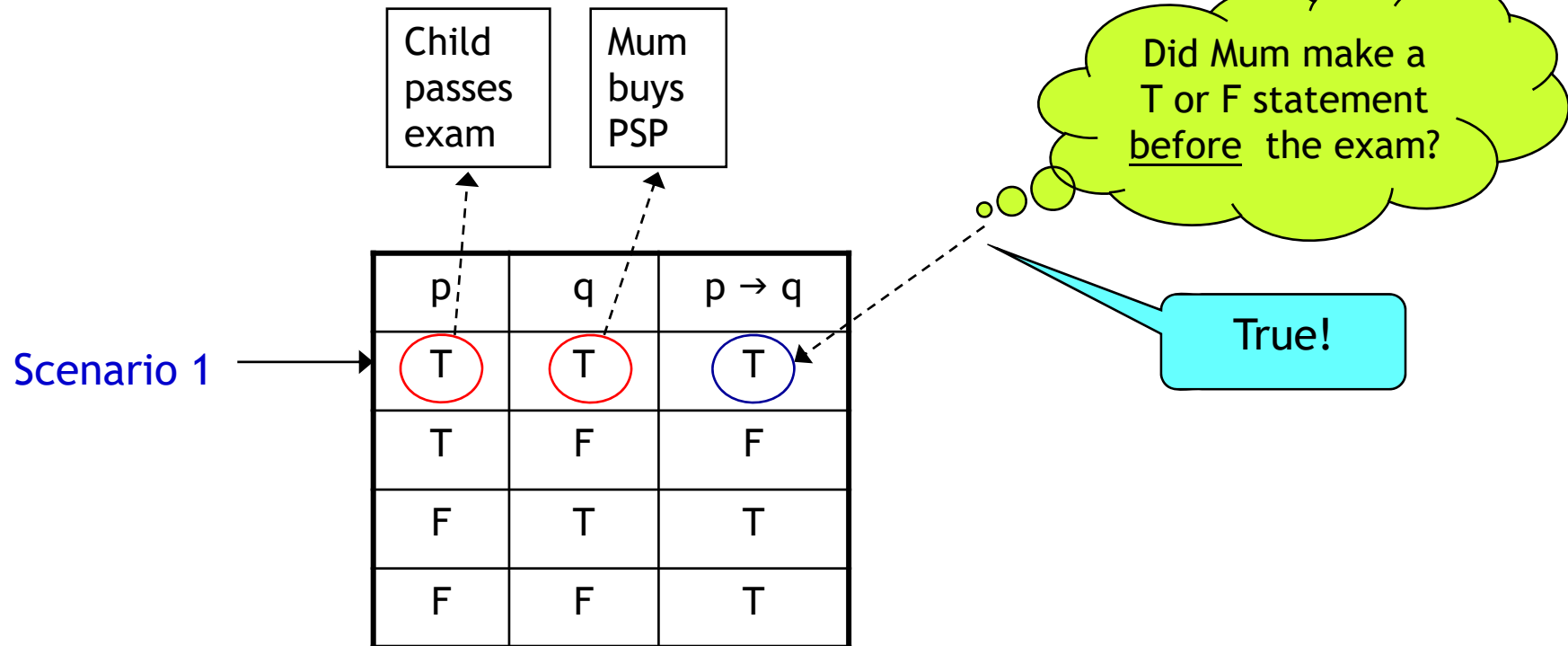
Observations

- The conditional $p \rightarrow q$ is false only when p is true and q is false.
- If p is false, then the conditional $p \rightarrow q$ is true regardless of the truth value of q . This is called true by default.

Rationale Behind the Truth Table

To understand the truth table for conditional statements, consider this example :

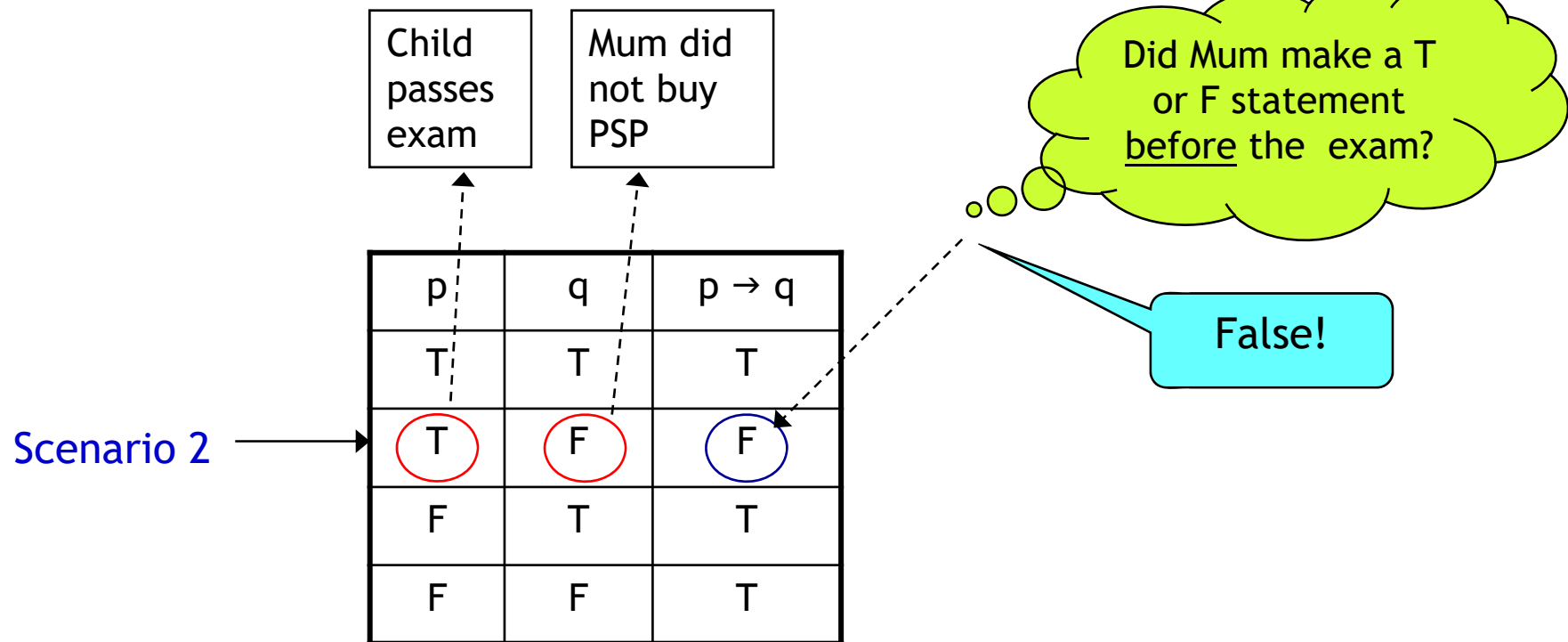
- Suppose a mother made this statement to her child before exam :
“If you pass your exam, I will buy you a PSP.”
- After exam, consider the following scenarios :



Rationale Behind the Truth Table (Cont'd)

To understand the truth table for conditional statements, consider this example :

- Suppose a mother made this statement to her child before exam :
“If you pass your exam, I will buy you a PSP.”
- After exam, consider the following scenarios :



Rationale Behind the Truth Table (Cont'd)

To understand the truth table for conditional statements, consider this example :

- Suppose a mother made this statement to her child before exam :
"If you pass your exam, I will buy you a PSP."

- After exam, consider the following scenarios :

Child did not
even pass
exam

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Scenario 3

Scenario 4

Did Mum make a T or
F statement before
the exam?

Can't be sure ...

The child did not even
pass exam ... so her
statement is not put
to the test

Then we will assume
Mum is innocent until
proven guilty!

We call this "True by
default"!

Order of Operations

In statements that involve $\sim$, $\wedge$, $\vee$ and $\rightarrow$, the **order of operations** follow these rules :

1. Perform $\sim$ first
2. $\wedge$ and $\vee$ are co-equal
3. $\rightarrow$
4. The use of brackets () will supersede all other operations.

Example (Truth Table involving $\rightarrow$)

Step 1 : Set up Rows

Step 2 : Strategize

Construct the truth table for $(x \rightarrow y) \rightarrow (x \wedge y)$.

Solution

No. of inputs = 2 No. of rows = $2^2 = 4$

Inputs				
x	y	$x \rightarrow y$	$x \wedge y$	$(x \rightarrow y) \rightarrow (x \wedge y)$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	F
F	F	T	F	F

Step 1 : Pick up the cases where hypothesis is “T” and conclusion is “F”. The output should be a “F”.

Step 2 : For all other rows, the output are “T”.

Exercise (Logical Equivalence Involving $\rightarrow$)

Prove or disprove the logical equivalence of $p \vee \sim q \rightarrow \sim p$ and $p \rightarrow p \vee \sim q$.

Solution

No. of inputs = 2 No. of rows = $2^2 = 4$

Inputs					(1)	(2)
p	q	$\sim q$	$p \vee \sim q$	$\sim p$	$p \vee \sim q \rightarrow \sim p$	$p \rightarrow p \vee \sim q$
T	T	F	T	F	F	T
T	F	T	T	F	F	T
F	T	F	F	T	T	T
F	F	T	T	T	T	T

Note the order of operations

Since rows 1 & 2 are different, the 2 statements are not equivalent.

Definition of Bi-conditional Statement (For Information - Not Tested)

Definition

Let p and q be statement variables.

A bi-conditional $p \leftrightarrow q$, read as “ p if and only if q ” or “ p iff q ” means “if p then q ” and “if q then p ”.

The truth table is as follows :

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Observation

- The conditional $p \leftrightarrow q$ is true when p and q have the same truth values.
- The conditional $p \leftrightarrow q$ is false when p and q have opposite truth values.

Part 3 Conditional Statement and Its Variations

Section 3

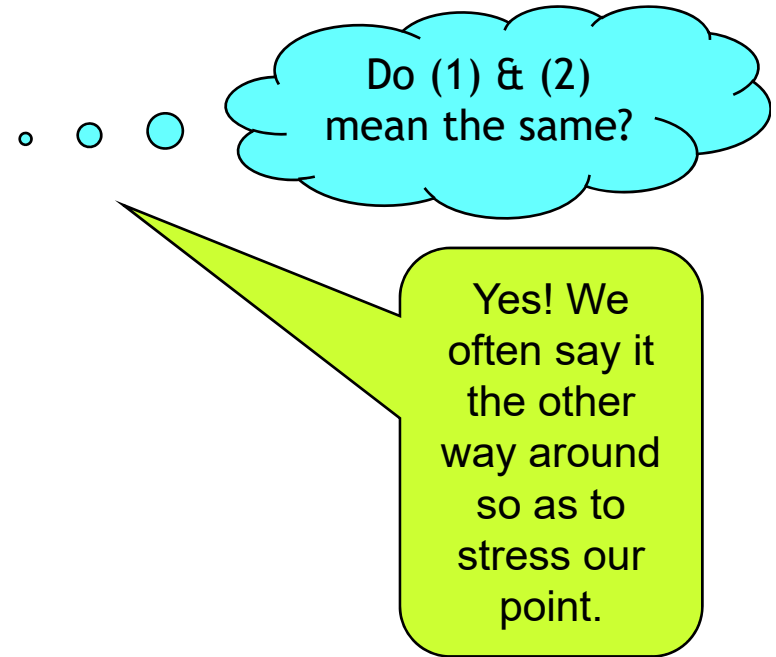
Contrapositive, Converse & Inverse Statements

- Introduction
- Definition
- Logical Equivalence of the Variations
- Example and Exercise - From Conditional to Its Variations

Introduction

Consider the following statements :

- (1) If you are sick, then you stay home.
- (2) If you do not stay home, then you are not sick.



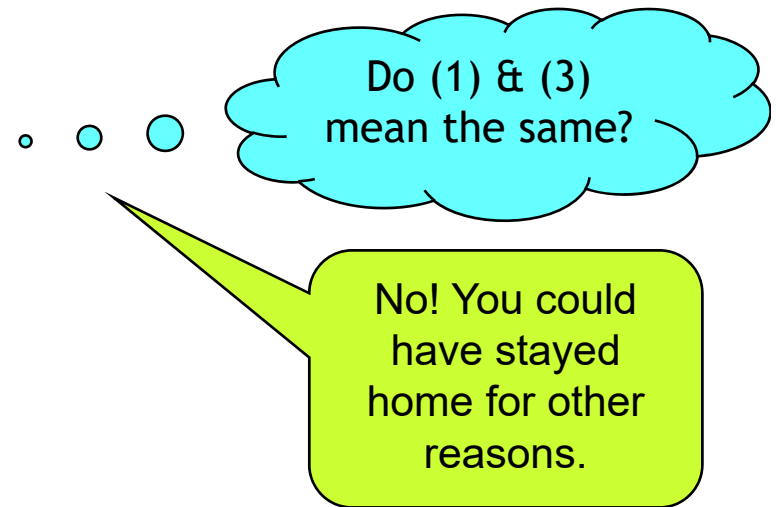
Keep in mind that ...

1. (1) and (2) have same meaning.

Introduction (Cont'd)

Now add in a 3rd statement :

- (1) If you are sick, then you stay home.
- (2) If you do not stay home, then you are not sick.
- (3) If you stay home, then you are sick.



Keep in mind that ...

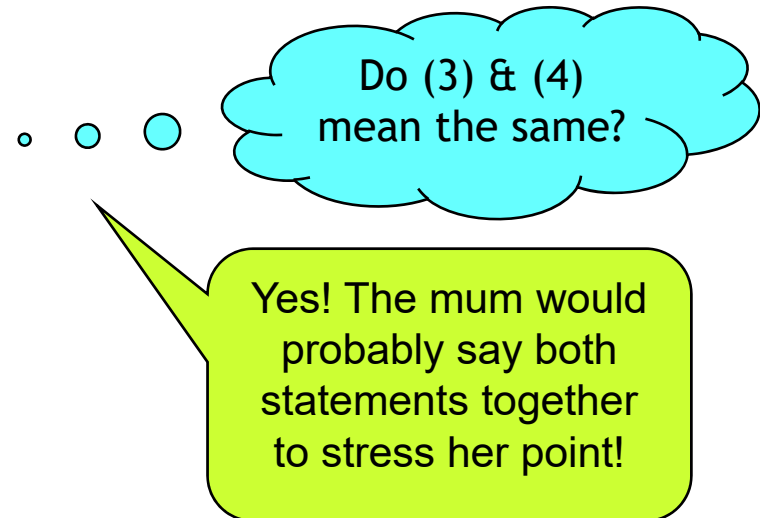
- 1. (1) and (2) have same meaning.
- 2. (1) and (3) have different meaning.

Introduction (Cont'd)

Imagine what a mum would say to her teenage child when she gets really fed up with him/her.

(3) If you stay home, then you are sick.

(4) If you are not sick, then you do not stay home.



Keep in mind that ...

3. (3) and (4) have same meaning.

Introduction (Cont'd)

Put all the statements together :

- (1) If you are sick, then you stay home.
- (2) If you do not stay home, then you are not sick.
- (3) If you stay home, then you are sick.
- (4) If you are not sick, then you do not stay home.

Keep in mind that ...

- 1. (1) and (2) have same meaning.
- 2. (1) and (3) have different meaning.
- 3. (3) and (4) have same meaning.

Introduction (Cont'd)

Change the statements into symbolic form and we get the following :

(1) If $\underbrace{\text{you are sick}}_p$, then $\underbrace{\text{you stay home}}_q$.

Conditional

(2) If $\underbrace{\text{you do not stay home}}_{\sim q}$, then $\underbrace{\text{you are not sick}}_{\sim p}$.

Contrapositive

(3) If $\underbrace{\text{you stay home}}_q$, then $\underbrace{\text{you are sick}}_p$.

Converse

(4) If $\underbrace{\text{you are not sick}}_{\sim p}$, then $\underbrace{\text{you do not stay home}}_{\sim q}$.

Inverse

1. A conditional statement $p \rightarrow q$

- is logically equivalent to its contrapositive.
- is not logically equivalent to its converse or inverse.

2. However, the converse and inverse are logically equivalent.

Definition

Definition

The contrapositive of $p \rightarrow q$ is $\sim q \rightarrow \sim p$

The converse of $p \rightarrow q$ is $q \rightarrow p$

The inverse of $p \rightarrow q$ is $\sim p \rightarrow \sim q$

So...

- Converse is arrived by ...
 - switching the hypothesis with the conclusion.
- Inverse is arrived by ...
 - replacing the hypothesis and the conclusion with their negation.
- Contrapositive is arrived by ...
 - switching the hypothesis with the conclusion;
 - and then negating both sides.

Logical Equivalence of the Variations

You can
prove
these
yourself!

The truth table of the 4 compound statements are as follows :

Inputs		Conditional	Contrapositive	Converse	Inverse
p	q	$p \rightarrow q$	$\sim q \rightarrow \sim p$	$q \rightarrow p$	$\sim p \rightarrow \sim q$
T	T	T	T	T	T
T	F	F	F	T	T
F	T	T	T	F	F
F	F	T	T	T	T



1. A conditional statement $p \rightarrow q$

- is logically equivalent to its contrapositive.
- is not logically equivalent to its converse or inverse.

2. However, the converse and inverse are logically equivalent.

Example (From Conditional To Its Variations)

Write the contrapositive, converse and inverse of the following conditional $p \rightarrow q$:

“If you drink, then don't drive.”

$\underbrace{\hspace{2cm}}_p \qquad \underbrace{\hspace{2cm}}_q$

Solution :

Contrapositive	$\sim q \rightarrow \sim p$	If <u>you drive</u> , then <u>you don't drink</u> . $\underbrace{\hspace{2cm}}_{\sim q} \qquad \underbrace{\hspace{2cm}}_{\sim p}$
Converse	$q \rightarrow p$	If <u>you don't drive</u> then <u>you drink</u> . $\underbrace{\hspace{2cm}}_q \qquad \underbrace{\hspace{2cm}}_p$
Inverse	$\sim p \rightarrow \sim q$	If <u>you don't drink</u> then <u>you drive</u> . $\underbrace{\hspace{2cm}}_{\sim p} \qquad \underbrace{\hspace{2cm}}_{\sim q}$

Exercise (From Conditional To Its Variations)

Write the contrapositive, converse and inverse of the following conditional $p \rightarrow q$:

“If you are hungry, then you eat.”

$\underbrace{\hspace{10em}}_p \qquad \underbrace{\hspace{10em}}_q$

Solution :

Contrapositive	$\sim q \rightarrow \sim p$	If <u>you don't eat</u> , then <u>you are not hungry</u> . $\underbrace{\hspace{10em}}_{\sim q} \qquad \underbrace{\hspace{10em}}_{\sim p}$
Converse	$q \rightarrow p$	If <u>you eat</u> , then <u>you are hungry</u> . $\underbrace{\hspace{10em}}_q \qquad \underbrace{\hspace{10em}}_p$
Inverse	$\sim p \rightarrow \sim q$	If <u>you are not hungry</u> , then <u>you don't eat</u> . $\underbrace{\hspace{10em}}_{\sim p} \qquad \underbrace{\hspace{10em}}_{\sim q}$

Summary ...

- Conditional
 - Need to know how to construct the truth tables
 - Need to know Contrapositive, Converse & Inverse
- Need to know that
 - Conditional $\equiv$ Contrapositive
 - Converse $\equiv$ Inverse
 - But Conditional is not logically equivalent to Converse

Application : Digital Logic Circuits

Binary Digits (Bits)

- Combine operations of electronic switches with logical connectives
- Replace T and F in signals of electronic switches with symbols 1 and 0 (known as *bits*)

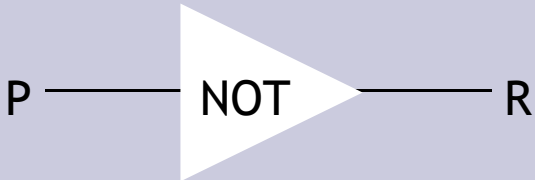
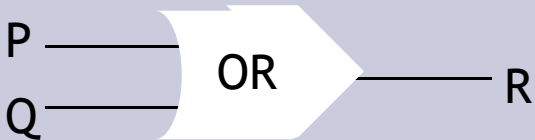
Logic Gates

- Combine signal bits (1 and 0) to form simple circuits called *gates*.
- NOT, AND, OR gates correspond to $\sim$, $\wedge$, $\vee$ in logical connectives.

Logic Circuits

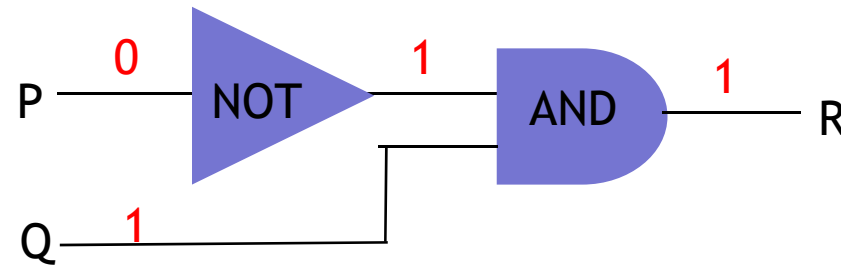
- Gates can then be combined into circuits in a variety of ways.

Application : Digital Logic Circuits

Type of Gate	Symbolic Representation	Action																		
NOT		<table><tr><th colspan="2">Input</th><th>Output</th></tr><tr><th>P</th><th>R</th><th></th></tr><tr><td>1</td><td>0</td><td></td></tr><tr><td>0</td><td>1</td><td></td></tr></table>	Input		Output	P	R		1	0		0	1							
Input		Output																		
P	R																			
1	0																			
0	1																			
AND		<table><tr><th colspan="2">Input</th><th>Output</th></tr><tr><th>P</th><th>Q</th><th>R</th></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td></tr></table>	Input		Output	P	Q	R	1	1	1	1	0	0	0	1	0	0	0	0
Input		Output																		
P	Q	R																		
1	1	1																		
1	0	0																		
0	1	0																		
0	0	0																		
OR		<table><tr><th colspan="2">Input</th><th>Output</th></tr><tr><th>P</th><th>Q</th><th>R</th></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr></table>	Input		Output	P	Q	R	1	1	1	1	0	1	0	1	1	0	0	0
Input		Output																		
P	Q	R																		
1	1	1																		
1	0	1																		
0	1	1																		
0	0	0																		

Application : Digital Logic Circuits

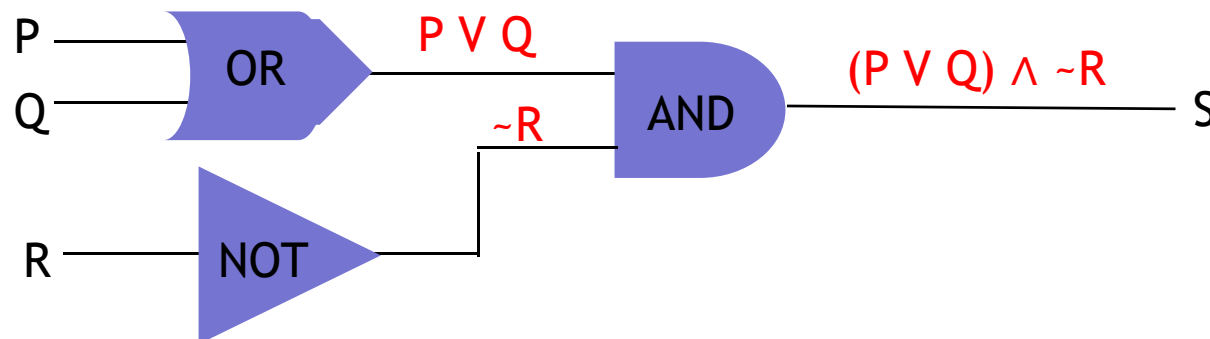
Determining Output for a Given Input



Boolean Expression Corresponding to a Circuit

A variable that takes one of only two values is called a Boolean variable.

An expression made up of Boolean variables and connectives $\sim$, $\wedge$, $\vee$ is called a Boolean expression.



Application : Number Systems & Circuits for Addition

Convert Decimal to Binary Numbers

Humans use decimal notation (base 10) for convenience as we happen to have 10 fingers. On the other hand, computers use binary notation (base 2) as modern electronics use two states (0 and 1) for their signals.

For example, the number 22 is represented as follows :

Decimal Notation : $22 = 2 \times 10^1 + 2 \times 10^0$

Place Value	10^4	10^3	10^2	10^1	10^0
Decimal Digits	0	0	0	2	2

So we write 22_{10} .

Binary Notation : $22 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$

Place Value	2^4	2^3	2^2	2^1	2^0
Decimal Digits	1	0	1	1	0

So we write 10110_2 .

Application : Number Systems & Circuits for Addition

Addition in Binary Notation (P+Q)

When adding 2 binary digits (P+Q), the following scenarios can happen :

$$\begin{array}{r} 0_2 \\ + 0_2 \\ \hline 00_2 \end{array}$$

$$\begin{array}{r} 1_2 \\ + 0_2 \\ \hline 01_2 \end{array}$$

$$\begin{array}{r} 0_2 \\ + 1_2 \\ \hline 01_2 \end{array}$$

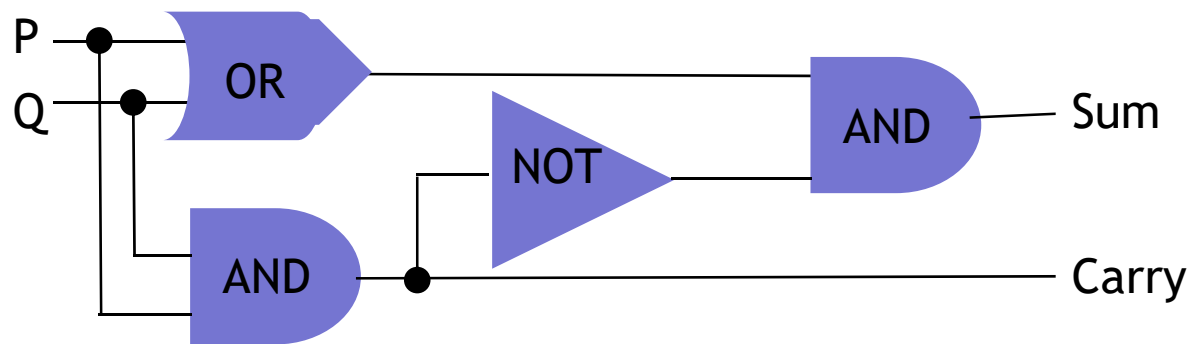
$$\begin{array}{r} 1_2 \\ + 1_2 \\ \hline 10_2 \end{array}$$

In the computer, digital logic circuits will be used to perform such additions.

Application : Number Systems & Circuits for Addition

Half-Adder (Add 2 Binary Digits : P+Q)

- A circuit is used to add 2 binary inputs P and Q.
- The circuit produces 2 outputs - left digit (**carry**) & right digit (**sum**).
- The carry output is 1 if P=1 and Q=1. Otherwise, it is 0.
 - So, the carry output corresponds to **AND**-gate circuit ($P \wedge Q$).
- The sum output is 1 if P=1 or Q=1, but not both=1. Otherwise, it is 0.
 - So, the sum output corresponds to *exclusive or* ($P \vee Q \wedge \sim(P \wedge Q)$).



P	Q	Carry	Sum
1	1	1	0
1	0	0	1
0	1	0	1
0	0	0	0

Application : Number Systems & Circuits for Addition

Addition in Binary Notation (P+Q+R)

What happens if the following scenarios arises :

$$\begin{array}{r} 111_2 \\ + 11_2 \\ \hline 110_2 \end{array}$$

← carry over

How does the computer handle the sum of the left column where there are actually three binary digits?

$$\begin{array}{r} 1_2 \\ 1_2 \\ + 1_2 \\ \hline 11_2 \end{array}$$

Application : Number Systems & Circuits for Addition

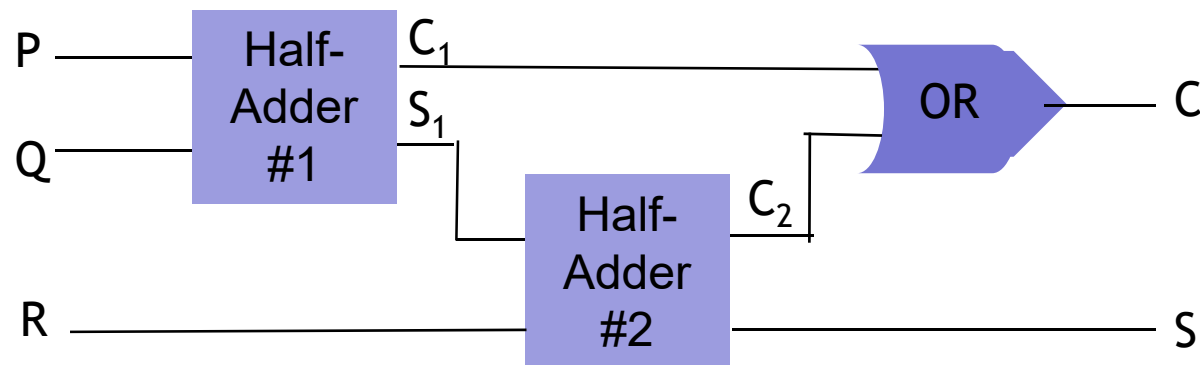
Full-Adder (Add 3 Binary Digits P+Q+R)

To add three binary numbers,

$$\begin{array}{r} P \\ Q \\ + R \\ \hline CS \end{array}$$

P is first added to Q. The result is then added to R.

The circuit used is to do that is a full-adder, which is based on half-adders, as shown below :



P	Q	R	Carry	Sum
1	1	1	1	1
1	1	0	1	0
1	0	1	1	0
1	0	0	0	1
0	1	1	1	0
0	1	0	0	1
0	0	1	0	1
0	0	0	0	0

*****END OF TOPIC*****